Appendix 2 Benchmark variables and coding rules

The normalized capability layer contains 48 variables for each of the 20 providers. The variables record feature states, while evidence strength is stored separately.

Table A2-1 Capability-state codebook

State	Code	Definition	Positive-capability value	Included as current?
Current	CUR	Feature was evidenced as current at the research cut-off.	1.0	Yes for positive features
Restricted / partial	PART	Feature existed with material limits or only part of the claim was supported.	0.5	Partly
Announced / future	FUT	Feature was announced or planned but not evidenced as current.	0.25	No
Explicitly unavailable	NO	Evidence stated that the positive feature was unavailable.	0	No
Unknown / no evidence	UNK	No sufficiently specific evidence was found.	Blank	Excluded

Source: author's capability-state codebook.

The sales or quote gate is a reverse-polarity variable. A current gate reduces accessibility, while an explicit absence of the gate is favorable. Unknown values remain unknown.

Table A2-2 Main composite measures

Measure	Components
Technical maturity	Equal-weight mean of Security, Lifecycle, Orchestration, Storage, Networking, Reliability, Monitoring, Pricing, Partnerships and Availability.
Market momentum	Equal-weight mean of Capital/execution, NVIDIA ecosystem, Customer validation, Partner depth, Availability/capacity signal and Brand visibility.
Accessibility	Equal-weight mean of Sign-up/account access, Authenticated console, Provisioning ease, Pricing transparency and Developer/self-service orientation.
Enterprise readiness	Equal-weight mean of Security, Lifecycle, Reliability, Monitoring, Enterprise focus, Customer validation and Partner depth.
Strategic relevance	Equal-weight mean of Technical maturity, Market momentum and Enterprise readiness.

Source: author's normalized comparison methodology.

Table A2-3 Standardized capability variables

Feature group	Variable code	Display label	Polarity	Operational definition
GPU portfolio	GPU_H100	NVIDIA H100	Positive	Evidence of current, partial or announced access to NVIDIA H100 GPUs.
GPU portfolio	GPU_H200	NVIDIA H200	Positive	Evidence of current, partial or announced access to NVIDIA H200 GPUs.
GPU portfolio	GPU_B200	NVIDIA B200	Positive	Evidence of current, partial or announced access to NVIDIA B200 GPUs.
GPU portfolio	GPU_B300	NVIDIA B300	Positive	Evidence of current, partial or announced access to NVIDIA B300 GPUs.
GPU portfolio	GPU_GB200	NVIDIA GB200	Positive	Evidence of current, partial or announced access to NVIDIA GB200 systems.
GPU portfolio	GPU_GB300	NVIDIA GB300	Positive	Evidence of current, partial or announced access to NVIDIA GB300 systems.
GPU portfolio	GPU_MI300X	AMD MI300X	Positive	Evidence of current, partial or announced access to AMD MI300X GPUs.
GPU portfolio	GPU_L40S	NVIDIA L40S	Positive	Evidence of current, partial or announced access to NVIDIA L40S GPUs.
GPU portfolio	GPU_A100	NVIDIA A100	Positive	Evidence of current, partial or announced access to NVIDIA A100 GPUs.
GPU portfolio	GPU_RTX_PRO_6000	NVIDIA RTX PRO 6000	Positive	Evidence of current, partial or announced access to NVIDIA RTX PRO 6000 GPUs.
Deployment model	Bare_metal	Bare metal	Positive	Dedicated physical GPU server access without a customer-facing virtual-machine layer.
Deployment model	VM_instances	VM instances	Positive	Customer-configurable virtual machines or instances with GPU resources.
Deployment model	Kubernetes	Kubernetes	Positive	Kubernetes-based deployment or managed Kubernetes support.
Deployment model	Slurm	Slurm	Positive	Access to or support for the Slurm workload scheduler.
Deployment model	Managed_clusters	Managed clusters	Positive	Provider-managed multi-node GPU cluster service.
Deployment model	Serverless_containers	Serverless / containers	Positive	Container or serverless compute service for AI workloads.

Deployment model	Marketplace_aggregator	Marketplace / aggregator	Positive	Marketplace or aggregation service that mediates access to external or multi-provider GPU capacity.
AI platform	Notebooks_studios	Notebooks / studios	Positive	Hosted notebook, studio, workspace or similar interactive development environment.
AI platform	Training_jobs	Training jobs	Positive	Managed or structured service for running model-training jobs.
AI platform	Inference_endpoints	Inference endpoints	Positive	Managed inference endpoint, model-serving service or model API.
AI platform	Model_catalog	Model catalog	Positive	Curated model catalog, model hub or model library.
AI platform	MLOps_observability	MLOps / observability	Positive	Experiment tracking, logs, metrics, monitoring or related MLOps functions.
Storage	Object_storage	Object storage	Positive	Object-storage service, including S3-compatible storage.
Storage	Block_storage	Block storage	Positive	Persistent block-volume service for workloads.
Storage	Shared_filesystem	Shared filesystem	Positive	Shared or network filesystem available across nodes or workloads.
Storage	High_perf_storage	High-performance storage	Positive	High-performance, parallel or local NVMe-oriented storage for data-intensive workloads.
Storage	Snapshots_backup	Snapshots / backup	Positive	Snapshot, backup or related storage-protection function.
Networking	InfiniBand	InfiniBand	Positive	InfiniBand network support for GPU workloads.
Networking	RoCE_RDMA	RoCE / RDMA	Positive	RoCE or RDMA support for low-latency data movement.
Networking	Ethernet	Ethernet	Positive	Ethernet networking offered or described for the service.
Networking	Multi_node_training	Multi-node training	Positive	Support or evidence for distributed training across more than one node.
Networking	NVLink_NVSwitch	NVLink / NVSwitch	Positive	NVLink, NVSwitch or NVL-based GPU interconnect support.
Access and pricing	Self_service_signup	Self-service signup	Positive	A user can create an account or enter the service without a prior sales process.
Access and pricing	Public_pricing	Public pricing	Positive	Pricing is available publicly for relevant services.

Access and pricing	Quote_only_sales	Sales / quote gate	Reverse	Relevant access or pricing requires sales contact, approval or a private quote.
Access and pricing	Free_trial_credits	Free trial / credits	Positive	Free trial, promotional credit or similar no-cost entry option.
Developer experience	Console_dashboard	Console / dashboard	Positive	Authenticated web console, portal or dashboard for service use.
Developer experience	CLI	CLI	Positive	Command-line interface for the provider service.
Developer experience	API_SDK	API / SDK	Positive	Public or customer-facing API or software development kit.
Developer experience	Terraform_IaC	Terraform / IaC	Positive	Terraform, OpenTofu or another infrastructure-as-code integration.
Developer experience	Support_docs	Documentation	Positive	Public or customer-accessible technical documentation and support guidance.
Security and compliance	SOC2	SOC 2	Positive	Public evidence of a SOC 2 report, attestation or certification claim.
Security and compliance	ISO27001	ISO/IEC 27001	Positive	Public evidence of ISO/IEC 27001 certification or a directly related claim.
Security and compliance	GDPR	GDPR	Positive	Public evidence of GDPR-related privacy or compliance support.
Security and compliance	HIPAA	HIPAA	Positive	Public evidence of HIPAA support, eligibility or a related compliance claim.
Security and compliance	Sovereign_cloud	Sovereign cloud	Positive	A sovereign-cloud, national-cloud or explicit data-residency service proposition.
Enterprise readiness	SLA	SLA	Positive	Published service-level agreement or relevant uptime/service commitment.
Enterprise readiness	Status_page	Public status page	Positive	Publicly accessible status page or incident communication surface.

Source: author's capability codebook.